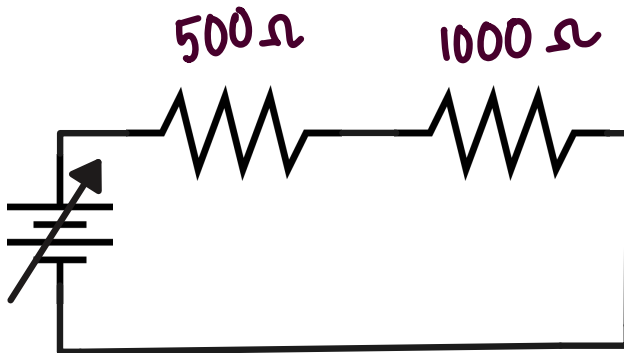


Assignment 1

Due Monday, January 26, 2026

1) A $500\ \Omega$ and a $1000\ \Omega$ resistor are connected in series, and both are rated to 0.25 W . The resistors are connected across a variable DC voltage supply. If you start at a low voltage and then turn up the voltage, which of the resistors will burn out first? At what voltage does this resistor burn out (assuming it does so right when it reaches its maximum rated power)?

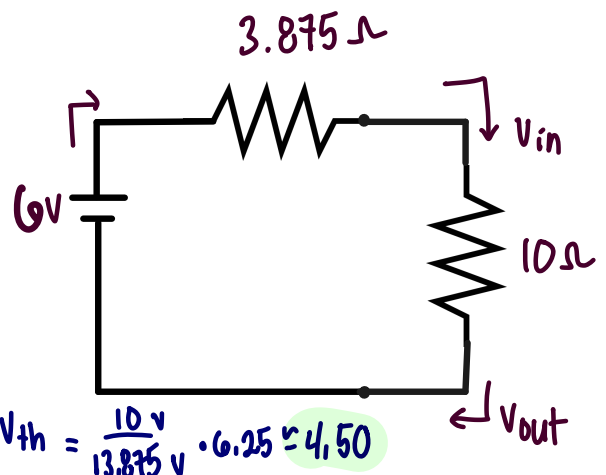
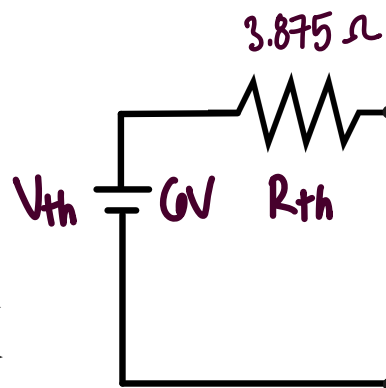
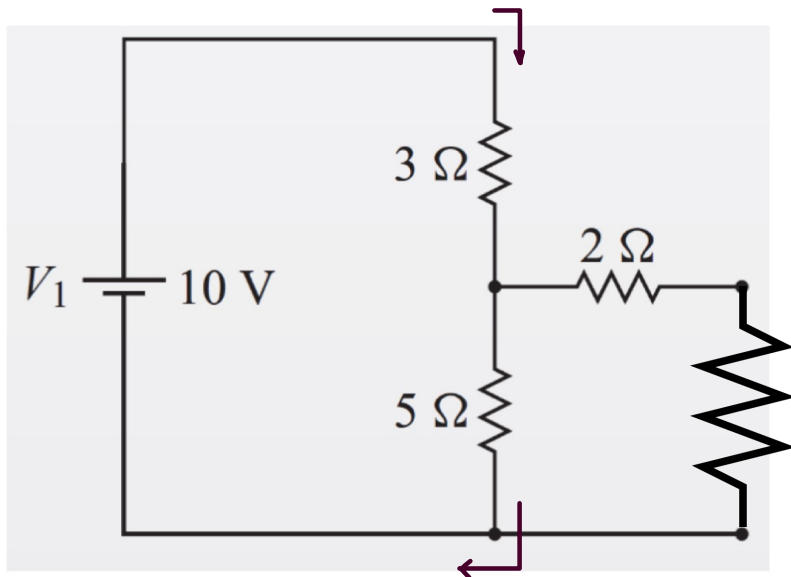


$$P = \frac{\Delta U}{\Delta t} = \frac{\Delta q V}{\Delta t} = IV = I^2 R$$

↳ w/ $P = I^2 R$ we see that the $1000\ \Omega$ resistor will burn out first

• Find V when $P = 0.25\text{ W}$: $V = IR$, $R_{\text{Tot}} = 1500$: $V = \sqrt{\frac{0.25\text{ W}}{1000}} \cdot 1500 \approx 23.7$

2) Find the Thevenin equivalent circuit for the circuit shown below. If a $10\ \Omega$ load resistor is placed across the output, what is voltage across the output?

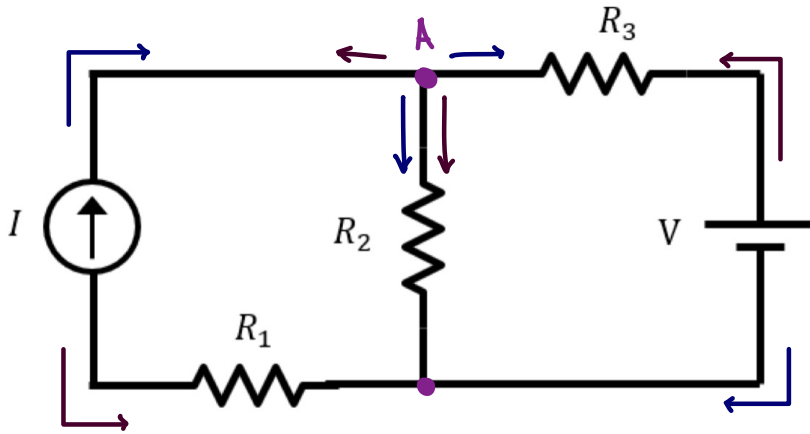


$$V_{th} = V \left(\frac{5\ \Omega}{8\ \Omega} \right) = \frac{50}{8} = 6.25\text{ V}$$

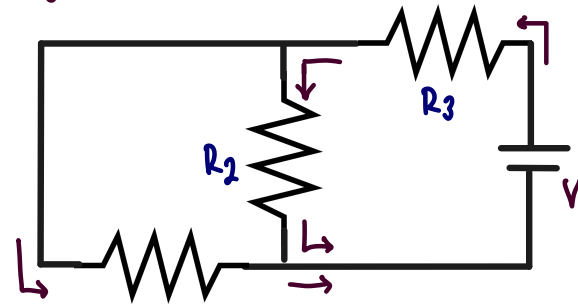
$$R_{\text{leq}} = \frac{3 \cdot 5}{3 + 5} = \frac{15}{8} = 1.875\ \Omega$$

↳ $R_{th} = R_{\text{leq}} + 2\ \Omega = 3.875\ \Omega \Rightarrow V_{out} = \frac{R_L}{R_{th} + R_L} \cdot V_{th} = \frac{10\ \Omega}{13.875\ \Omega} \cdot 6.25\text{ V} \approx 4.50\text{ V}$

3) A voltage source of voltage V , a current source with current I , and three resistors with resistances R_1 , R_2 , and R_3 are arranged into a circuit as shown below. What is the voltage across R_2 ? What is the current supplied by the voltage source?



voltage source only



$$V = IR \Rightarrow I = \frac{R_1}{R_2} + \frac{V_A - V}{R_3}$$

$$\hookrightarrow IR_2 R_3 = V_A R_3 + V_A R_2 - V R_2$$

$$IR_2 R_3 + V R_2 = V_A (R_2 + R_3)$$

$$\hookrightarrow V_{R_2} = \frac{R_2 (IR_3 + V)}{R_2 + R_3}$$

$$I_V = \frac{V - V_A}{R_3} = \frac{V - V_{R_2}}{R_3} = V - \frac{R_2 (IR_3 + V)}{R_2 + R_3}$$

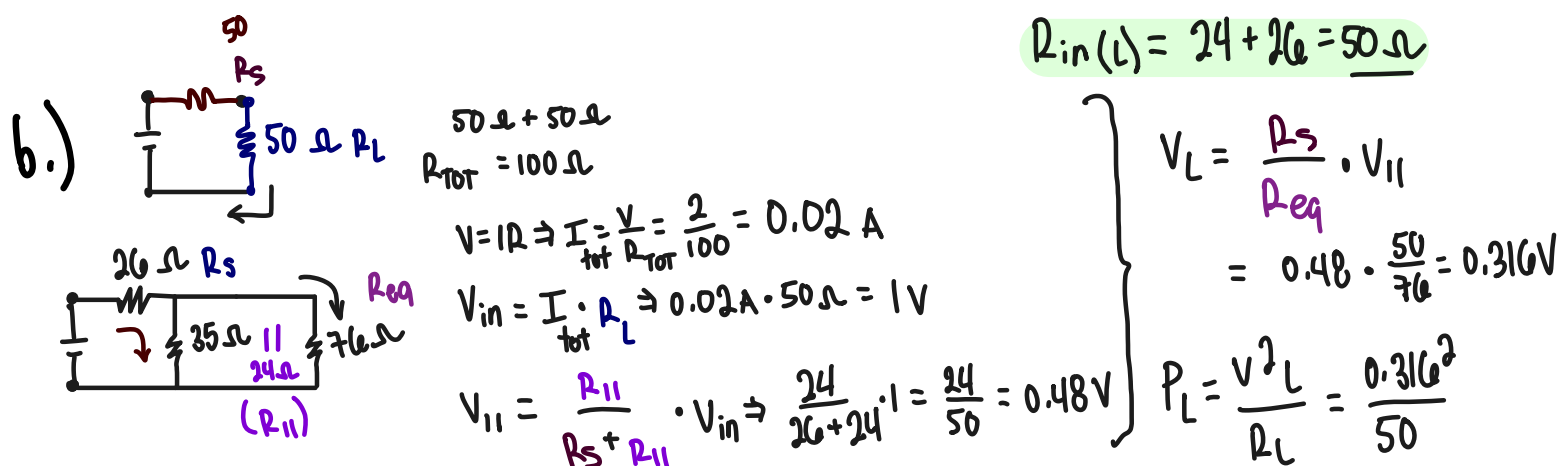
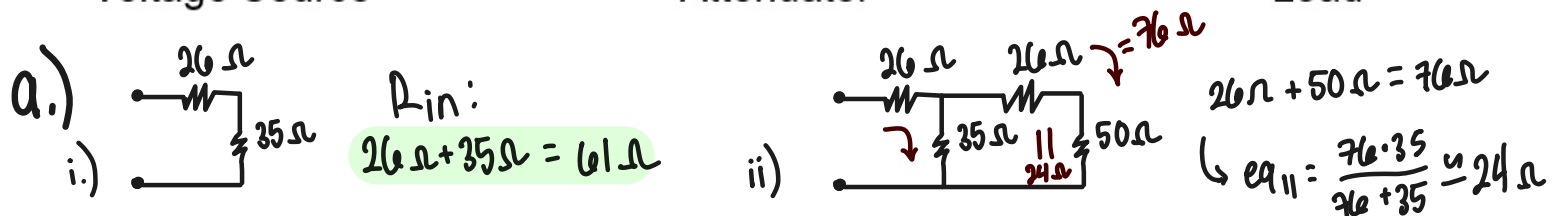
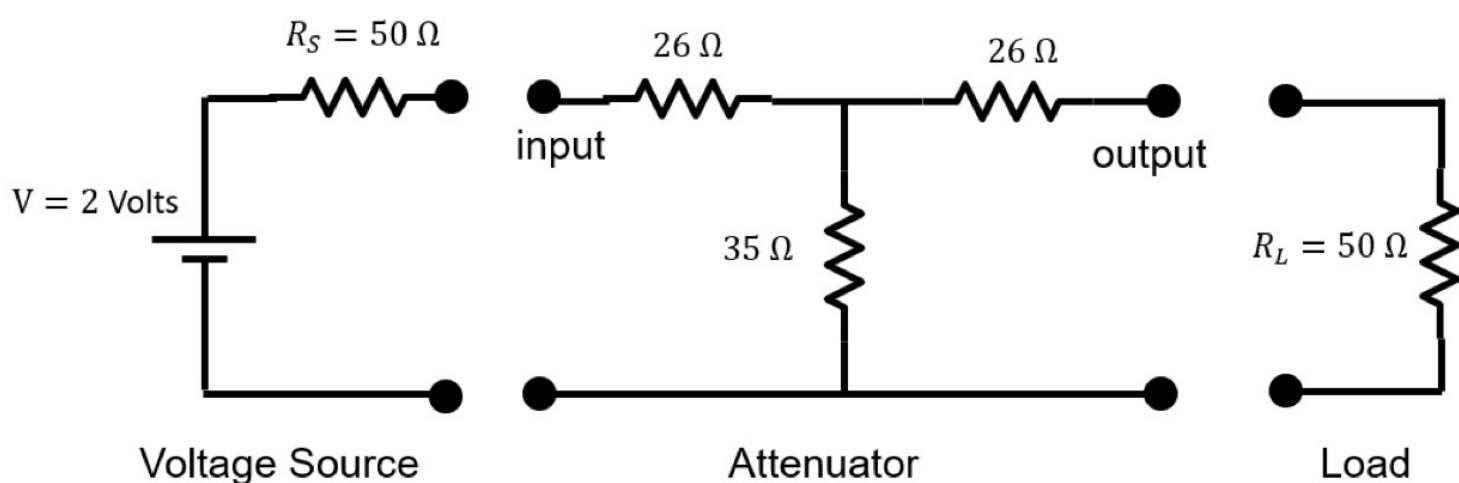
$$\hookrightarrow I_V = \frac{VR_2 + VR_3 - R_2 R_3 I - R_2 V}{R_3 (R_2 + R_3)} \Rightarrow \frac{\cancel{VR_3} - \cancel{IR_2 R_3}}{\cancel{R_3} (R_2 + R_3)} = \frac{V - IR_2}{R_2 + R_3}$$

4) A resistive attenuator circuit is drawn below. This circuit consists of 3 resistors with the values of the resistances given in the diagram.

a) Calculate the input resistance of the circuit if no load is attached to the output of the circuit. Also calculate the input resistance of the circuit if the load ($R_L = 50 \Omega$) is attached to the output of the circuit. (Assume the voltage source is not yet attached to the circuit.)

b) Calculate the power supplied to the load resistor if the voltage source with output resistance $R_S = 50 \Omega$ is connected to the input of the attenuator circuit and the load is attached to the output of the circuit.

c) Calculate the power supplied to the load resistor if it is connected directly to the output terminals of the voltage source. Then compute the ratio of the power you calculated in part (b) to the power you just calculated. This is called the attenuation factor.



c.) $V_{L_d} = V \cdot \frac{R_L}{R_S + R_L} = 2 \text{ V} \cdot \frac{50}{50 + 50} = 1 \text{ V}$

$$P_L(\text{direct}) = \frac{V_{L_d}^2}{R_L} = \frac{1^2}{50} = 20 \text{ mW}$$